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APPENDIX

A JUSTIFICATION FOR PROB-SUM ACTIVATION FUNCTION

To identify the most confident subpredicate among three possible candidates, we employ Eq. 5, which serves as a smooth and differentiable approximation of logical disjunction over fuzzy truth values. This formulation is grounded in fuzzy logic principles, particularly the product t -norm and De Morgan’s laws.

In fuzzy logic, disjunctions are typically modeled using t -conorms. One such t -conorm is derived from the product t -norm, where conjunction is approximated as $x \wedge y \approx xy$. Applying De Morgan’s law, the disjunction can be expressed as:

$$\mu_{x \vee y} = 1 - (1 - \mu_x)(1 - \mu_y),$$

where μ_x and μ_y denote the fuzzy membership values of inputs x and y , respectively. This formulation naturally extends to three variables:

$$\mu_{x \vee y \vee z} = 1 - (1 - \mu_x)(1 - \mu_y)(1 - \mu_z).$$

Letting $\mu_x = x$, $\mu_y = y$, and $\mu_z = z$, we expand the expression:

$$\begin{aligned} f(x, y, z) &= \mu_{x \vee y \vee z} \\ &= 1 - (1 - x)(1 - y)(1 - z) \\ &= 1 - (1 - x - y - z + xy + xz + yz - xyz) \\ &= x + y + z - xy - xz - yz + xyz. \end{aligned}$$

This derivation confirms that $f(x, y, z)$ corresponds exactly to the fuzzy disjunction defined by the product t -conorm extended to three variables. The function captures the cumulative confidence across all subpredicates while attenuating redundancy via subtraction of pairwise interactions and reinforcing agreement via the $+xyz$ term.

By design, this function produces higher values when at least one input is close to 1.0, thereby emphasizing the most confident subpredicate. Since our objective is to identify the subpredicate with the highest contribution to the overall rule activation, this function offers a principled, differentiable, and interpretable measure rooted in fuzzy logic semantics.

B ATTENTION-BASED VS. PRODUCT-BASED OPERATORS

B.1 SOFTMIN ATTENTION VS. PRODUCT t -NORM

Lemma 1. Let $\mathbf{b} = (b_1, b_2, \dots, b_m) \in \mathbb{R}^m$ and let $b_{\min} = \min_i b_i$. Then for any $\beta > 0$, the softmin operator

$$T_{\text{attention}}(\beta) = \sum_{i=1}^m \alpha_i(\beta) \cdot b_i, \quad \text{where } \alpha_i(\beta) = \frac{e^{-\beta b_i}}{\sum_{j=1}^m e^{-\beta b_j}},$$

satisfies:

$$T_{\text{attention}}(\beta) \geq b_{\min},$$

with equality if and only if $b_i = b_{\min}$ for all i or in the limit $\beta \rightarrow \infty$.

Proof. Each $\alpha_i(\beta)$ is strictly positive and the weights form a convex combination: $\sum_i \alpha_i(\beta) = 1$ and $\alpha_i(\beta) > 0$. Let $b_{\min} = b_k$ for some index k . Then:

$$T_{\text{attention}}(\beta) = \sum_{i=1}^m \alpha_i(\beta) \cdot b_i = b_{\min} \sum_{i=1}^m \alpha_i(\beta) + \sum_{i=1}^m \alpha_i(\beta) \cdot (b_i - b_{\min}).$$

Since each $b_i - b_{\min} \geq 0$ and at least one $b_i > b_{\min}$ (unless all values are equal), the second term is strictly positive. Therefore:

$$T_{\text{attention}}(\beta) > b_{\min} \quad \text{unless } b_i = b_{\min} \text{ for all } i.$$

In the limit $\beta \rightarrow \infty$, the softmin attention converges to the minimum:

$$\lim_{\beta \rightarrow \infty} T_{\text{attention}}(\beta) = b_{\min}.$$

□

Theorem 1. Let $\mathbf{b} = (b_1, b_2, \dots, b_m) \in (0, 1]^m$ be a vector of probabilistic values, and let $b_{\min} = \min_i b_i$. Define the following:

- *Softmin attention:*

$$T_{\text{attention}}(\beta) = \sum_{i=1}^m \alpha_i(\beta) \cdot b_i \quad \text{where } \alpha_i(\beta) = \frac{e^{-\beta b_i}}{\sum_{j=1}^m e^{-\beta b_j}}$$

- *Product t-norm:*

$$T_{\text{prod}} = \prod_{i=1}^m b_i$$

Then, there exists a finite $\beta^* > 0$ such that the approximation error of the softmin attention operator is strictly less than that of the product t-norm:

$$\epsilon_{\text{attention}}(\beta^*) = T_{\text{attention}}(\beta^*) - b_{\min} < b_{\min} - T_{\text{prod}} = \epsilon_{\text{prod}}.$$

Proof. Since all $b_i \in (0, 1]$, the product t-norm satisfies $T_{\text{prod}} < b_{\min}$ (unless all $b_i = 1$), so:

$$\epsilon_{\text{prod}} = b_{\min} - T_{\text{prod}} > 0.$$

By the Lemma, $T_{\text{attention}}(\beta) > b_{\min}$ for all finite β , and:

$$\lim_{\beta \rightarrow \infty} T_{\text{attention}}(\beta) = b_{\min} \quad \Rightarrow \quad \lim_{\beta \rightarrow \infty} \epsilon_{\text{attention}}(\beta) = 0.$$

Because $T_{\text{attention}}(\beta)$ is a continuous and strictly decreasing function of β , and since:

$$\epsilon_{\text{prod}} > 0 = \lim_{\beta \rightarrow \infty} \epsilon_{\text{attention}}(\beta),$$

then by the intermediate value theorem, there exists a finite $\beta^* > 0$ such that:

$$\epsilon_{\text{attention}}(\beta^*) < \epsilon_{\text{prod}} \quad \Leftrightarrow \quad T_{\text{attention}}(\beta^*) < 2b_{\min} - T_{\text{prod}}.$$

Hence, softmin attention provides a strictly better approximation to min than the product t-norm for some finite value of β . □